

10. (a) Halogenoalkanes can be hydrolysed using water in a similar way to using aqueous sodium hydroxide.

A student carried out an experiment to investigate the rate of reaction for the hydrolysis of halogenoalkanes using water. The student used aqueous ethanol to dissolve the halogenoalkane and then added a few drops of aqueous silver nitrate. He timed how long it took to produce a precipitate. He obtained the results shown in the table.

Halogenoalkane	Time / s
1-chloropropane, C_3H_7Cl	300
1-bromopropane, C_3H_7Br	90
1-iodopropane, C_3H_7I	15

The student tried to explain these results and he looked on the internet to find the following data.

Bond	Bond enthalpy / kJ mol^{-1}
C—H	413
C—C	348
C—F	485
C—Cl	328
C—Br	276
C—I	240

Element	Electronegativity
chlorine	3.16
bromine	2.96
iodine	2.66
carbon	2.55



- (i) Use both sets of data to explain the student's results. Include an equation to show the hydrolysis reaction between a halogenoalkane and aqueous sodium hydroxide and name this type of reaction mechanism. [6 QER]

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- (ii) Write an **ionic** equation, including state symbols, for a reaction that produces a silver halide precipitate. [1]

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- (iii) Suggest a practical method by which the student could have obtained these results. [2]

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- (iv) Suggest the difference that the student would have observed in his experiments if he had not added ethanol to the water before adding the halogenoalkane. [1]

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- (b) Chlorofluorocarbons, CFCs, were historically used for a variety of commercial and domestic purposes but nowadays their use is very restricted.

- (i) Outline the adverse environmental effects of the use of CFCs.

You do not need to include any equations for the reactions involved. [4]

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- (ii) Use relevant data in part (a) to explain why hydrofluorocarbons, HFCs, have replaced CFCs in many of their uses. [2]

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